PRODUCT DATA SHEET

Athen Marine HYD HVI Series

Premium Hydraulic Fluids with High Viscosity Index for Wide Temperature Ranges.

Product Description

Athen Marine Hydraulic Oil HVI Series is a premium range of high viscosity index (HVI) hydraulic fluids formulated to deliver exceptional performance in marine and industrial hydraulic systems operating under wide temperature variations. These fluids are specifically designed for applications where minimal viscosity change is required despite fluctuating temperatures.

They meet the stringent performance requirements of modern hydraulic systems equipped with high-pressure, high-output pumps and the critical demands of precision components such as numerically controlled machine tools and servo valves with close clearances.

Available viscosity grades: 15 to 150

Features & Benefits

- Outstanding thermo-oxidative stability and low & high temperature performance allow for extended service life.
- They provide excellent anti-wear property, rust & corrosion protection, water separation & air-release properties, and hydrolytic stability to reduce breakdowns and help improve production capacity
- The high viscosity index assures equipment protection at cold start-up temperatures and protects system components at high operating temperatures.
- Compatible with multi-metals and sealing materials commonly used in hydraulic systems.

Applications

- Hydraulic and power transmission systems
- Hydraulic systems of excavators, cranes, and hydrostatic drives are subjected to the most severe outdoor operating conditions
- Hydraulic systems operating under high pressures and requiring a high degree of load-carrying capability and anti-wear protection.

PRODUCT DATA SHEET

Athen Marine

Athen Marine HYD HVI Series

Athen Marine HYD HVI Series								
Performance Level Meets/Exceeds the specifications								
ISO Viscosity Grades		**15**	**22**	**32**	**46**	**68**	**100**	**150**
DIN 51524 Part 3 HVLP		✓	✓	✓	✓	✓	✓	✓
ASTM D 6158 (HV)		✓	✓	✓	✓	✓	✓	✓
ISO 11158 HV		✓	✓	✓	✓	✓	✓	✓
Eaton E-FDGN-TB002-E		✓	✓	✓	✓	✓	✓	✓
GB 11118.1-2011 (L-HV)		✓	✓	✓	✓	✓	✓	✓
SAE MS 1004 (HV)		✓	✓	✓	✓	✓	✓	
SEB 181222			✓	✓	✓	✓	✓	
Parker (formerly Denison) HF-0, HF-1, HF-2				✓	✓	✓		
Bosch Rexroth RDE 90235				✓	✓	✓		
JCMAS P041 HK				✓	✓			
Fives CINCINNATI (Former MAG IAS, LLC)				P-68	P-70	P-69		
Typical Properties								
Test Parameters		**15**	**22**	**32**	**46**	**68**	**100**	**150**
Viscosity @ 40 °C, cSt	D 445	15	22	32	46	68	100	150
Viscosity Index	D 2270	145	145	145	145	145	145	145
Flash Point, °C	D 92	180	190	220	220	230	240	250
Pour Point, °C	D 97	-42	-42	-39	-36	-33	-27	-27
Density @ 15°C, Kg/l	D 1298	0.850	0.852	0.855	0.857	0.860	0.862	0.865
Rust Test	D 665A/B	Pass	Pass	Pass	Pass	Pass	Pass	Pass
Emulsion Test, 54 °C, 30 minutes max	D 1401	Pass	Pass	Pass	Pass	Pass	-	-
Emulsion Test, 82 °C, 30 minutes max	D 1401	-	-	-	-	-	Pass	Pass
Foaming Characteristics, Seq-I, II, III, Stability (after 10 min. settling)	D 892	Nil	Nil	Nil	Nil	Nil	Nil	Nil
Turbine Oil Stability Test, hrs	D 943	5000+	5000+	5000+	5000+	5000+	4000+	4000+
FZG, fail load stage minimum	DIN ISO 14635-1	-	10	12	12	12	12	12

"Data provided in this PDS is based on standard tests under laboratory conditions and is indicative only. Minor variations that do not affect product performance are expected in normal manufacturing. This product should not be used for any purpose other than those expressly set out in this PDS"